

GMCM — Group Memory Coordination Module

OOF™ Origin Open Foundation™

Independent Methodological Authority

OriginID: OOF-OID-MEM-GMCM-2026-06-08-0002

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Memory Governance Intelligence Architecture (MGIA™)

Operational Layer: Group Memory Governance Layer

Governed Space: Group Memory Coordination Integrity

Category: AI & Interpretation

Subcategory: Memory Coordination Governance

Type: Group Memory Integrity Module

Parent Standard: Group Memory Integrity Standard (GMIS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 8 June 2026

Compatibility: OOF Methodology OS · Group Memory Integrity Standard (GMIS) ·

Shared Memory Integrity Module (SMIM) · Collective Cognition Integrity Standard (CCIS) ·

Multi-Agent Cognition Integrity Standard (MCIS) · Human-AI Cognition Integrity Standard (HAICS) ·

INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition System

Canonical Definition

Group Memory Coordination Module (GMCM) defines the structural conditions under which memory shared across multiple participants remains sufficiently coordinated, synchronized, aligned, traceable, accountable, and operationally valid to support collective continuity, collaboration, communication, and coordinated action.

GMCM governs group memory coordination.

The module ensures that participants maintain compatible memory states and shared operational understanding rather than diverging into disconnected or contradictory memory environments.

A system satisfies GMCM only if:

- memory coordination remains preservable
- synchronization remains assessable
- memory alignment remains detectable
- coordination failures remain identifiable
- collective memory remains governable
- coordination remains operationally valid

A group that cannot maintain coordinated memory states does not satisfy GMCM.

Module Operational Role

GMCM defines the memory-coordination governance layer of GMIS.

Its role is to preserve alignment of shared memory across participants.

Module Operational Space

GMCM governs:

- memory coordination
- memory synchronization
- collective alignment
- collaborative continuity
- shared operational understanding
- coordination accountability
- memory-state alignment
- coordination governance

The module applies wherever multiple participants depend on coordinated memory.

Module Function

The module applies wherever systems must preserve:

- synchronized memory states
- collective operational continuity
- aligned contextual understanding
- governance-valid coordination
- collaborative reliability
- coordinated decision support

Its function is to ensure that shared memory remains sufficiently coordinated to support collective operation.

Minimum Implementation Framework

1. Define the Memory Coordination Object

The organization must define which shared-memory environments require coordination governance.

This may include:

- project memory
 - team memory
 - Human-AI memory
-

- multi-agent memory
- organizational memory
- operational memory
- collaborative knowledge systems
- collective intelligence environments

2. Define Memory Coordination Conditions

The system must define the conditions under which coordination remains valid.
This includes:

- synchronization requirements
- alignment requirements
- continuity requirements
- accountability requirements
- coordination requirements
- governance-valid coordination conditions

3. Define Coordination Degradation Detection Logic

The system must define how coordination degradation is identified.
This may include:

- synchronization failures
- memory divergence
- context misalignment
- inconsistent memory states
- collaborative fragmentation
- coordination drift

4. Define Operational Response or Governance Logic

The system must define governance logic for coordination-integrity failures.
Governance response may include:

- synchronization activities
- alignment review
- memory reconciliation
- governance intervention
- escalation
- coordination restoration
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- synchronization events
- coordination activities

- alignment reviews
- governance interventions
- restoration procedures
- resulting memory states

A group-memory environment must not remain coordination-valid if materially significant coordination failures cannot be reconstructed, reviewed, or governed.

Use Case 1 — Human-AI Operations Center

Scenario

Humans and AI systems continuously collaborate while sharing operational knowledge, situational awareness, and mission-related memory.

Application

GMCM governs synchronization and coordination of shared memory across participants.

Result

The environment gains stronger operational alignment, improved coordination reliability, and reduced exposure to memory divergence.

Use Case 2 — Multi-Agent Autonomous Network

Scenario

Multiple autonomous agents coordinate activities through shared memory and collective operational knowledge.

Application

GMCM governs memory synchronization, collective alignment, and continuity of operational understanding.

Result

The network gains stronger coordination, improved collective reliability, and reduced exposure to coordination failures.

Canonical Closing Statement

Group Memory Coordination Module (GMCM) defines the structural conditions under which memory shared across multiple participants remains sufficiently coordinated, synchronized, aligned, traceable, accountable, and operationally valid to support collective continuity, collaboration, communication, and coordinated action.

**Collective operation cannot remain reliable if shared memory becomes uncoordinated.
Group memory coordination therefore becomes a foundational integrity condition of
collaborative intelligence systems.**

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>